

Stage 5 – main board only. DevKit USB–C programming and main–board USB–C power are mutually exclusive (approved Rev A policy).

No bare ESP32, 3V3 buck, EN/BOOT or programming circuit on this PCB. J1/J2 are the verified removable DevKit sockets.

J3 is universal E220–T22D interface: fit one E220–400T22D OR E220–900T22D. No socket/footprint is selected.

E220 M0/M1 use 10k pull–downs to GND (PROJECT DESIGN CHOICE): reset mode M1/M0=00. AUX is input–only, no pull–down.

OLED: GPIO21/SDA and GPIO22/SCL are signal–only reservations; OLED VCC and I2C pull–ups are NC/DNP in Rev A.

TUSB320 UFP/GPIO: PORT=GND, ADDR=NC, EN_N=GND. OUT1/Q1 gates TPS259630: Default/detach OFF; Medium/High ON.

USB4105–GF–A: all A/B VBUS, GND and shell contacts are explicit; D+ /D– are local NC nets, SBU pins NC, no ESP USB connection.

GPIO4 -> SN74AHCT1G125DBVR (OE=GND, V5) -> WS2812B–V5 DIN. No WS2812 footprint assigned at schematic stage.

VBUS_PRE -> U4 TLV70433DBVR -> PRE_GATE_3V3 powers TUSB320, TPD4S311 and all OUT1/EN pull–ups. MMSD4148T1G path is superseded/removed.

TPD4S311: C_CC1/C_CC2 from J4, RPD4S311/G2 tied to those nodes for dead–battery Rd; D3/D4 protected side to TUSB CC1/CC2. No permanent external 5.1k Rd.

D3 TPD1E10B06DPYR protects VBUS_PRE=PRE_GATE_3V3 + C10 1uF; VBIAS=C11 100nF/50V to GND. SBU/FLT NC.

TP1 5V_SYS; TP2 E220_VCC(5V_SYS); TP3 M0; TP4 M1; TP5 AUX; TP6 E220_RXD; TP7 E220_TXD. Test–point footprints/locations are deferred.

